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Bachelor Thesis in Physics
submitted by

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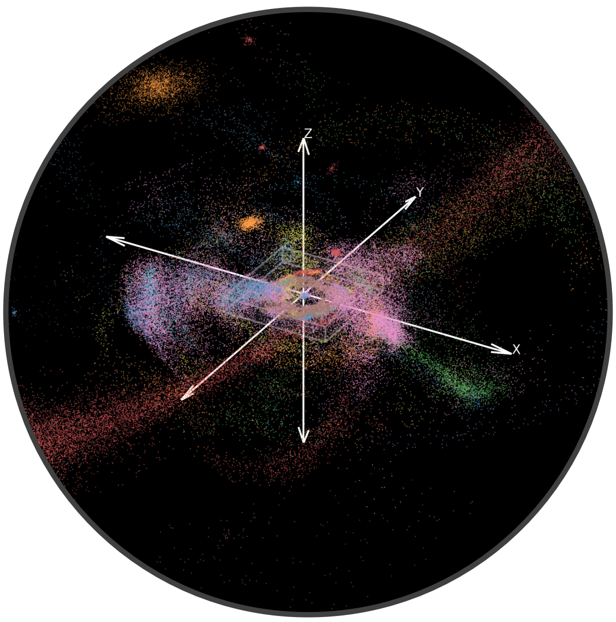
Classification and Analysis of Star-Forming Clusters in a Milky Way-Like Galaxy Simulation



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Abstract

The formation and dissolution of stellar clusters are fundamental processes in galactic evolution. However, identifying these structures, especially young clusters in the galactic disk, remains a challenge.

This thesis uses a pipeline of applying two general-purpose clustering algorithms (AstroLink and FuzzyCat) to a high-resolution cosmological simulation of a Milky Way-like galaxy from the NIHAO-UHD suite. We compare cluster identification using purely dynamical tagging (positions and velocities) and chemodynamical tagging (incorporating [Fe/H] and [O/H] abundances).

Our analysis reveals that the chemodynamical approach provides a far more complete census of stellar structures. It identifies $\sim 40\%$ more clusters overall and is twice as effective at locating young, star-forming clusters in the disk environment, where purely dynamical signatures are erased within a few orbital periods. From our simulations, we predict that the present-day ratio of native to pollution stars in a typical star-forming cluster in a Milky Way-like galaxy is about 4:1, providing a guideline for observational studies.

We conclude that chemical information is essential for an unbiased understanding of star formation, as purely kinematic surveys will fundamentally miss a significant fraction of stellar associations, leading to an incomplete picture of galaxy assembly.

Zusammenfassung

Die Entstehung und Auflösung von Sternhaufen sind fundamentale Prozesse in der Galaxienentwicklung. Die Identifikation dieser Strukturen, insbesondere von jungen Sternhaufen in der galaktischen Scheibe, bleibt jedoch eine Herausforderung.

In dieser Arbeit wird eine Pipeline vorgestellt, die zwei allgemeine Clustering-Algorithmen (AstroLink und FuzzyCat) auf eine hochauflösende, kosmologische Simulation einer Milchstraßen-ähnlichen Galaxie aus der NIHAO-UHD-Suite anwendet. Verglichen wird die Identifikation von Strukturen mittels rein dynamischer Merkmale (Positionen und Geschwindigkeiten) und einer chemisch-dynamischen Methode (unter Einbeziehung der [Fe/H] und [O/H]-Häufigkeiten).

Unsere Analyse zeigt, dass der chemisch-dynamische Ansatz ein deutlich vollständigeres Bild von stellaren Strukturen liefert. Dieser identifiziert insgesamt etwa 40% mehr Sternenhaufen, und ist doppelt so effektiv beim Auffinden junger, sternformender Gruppen im Umfeld der galaktischen Scheibe, wo sich rein dynamische Signaturen innerhalb weniger Umläufe verlieren. Ein zentrales Ergebnis für die beobachtende Astronomie ist, dass das theoretisch erwartete Verhältnis von im Sternenhaufen geborenen Sternen zu kontaminierenden Sternen etwa 4:1 beträgt.

Wir kommen insgesamt zu dem Schluss, dass chemische Informationen unerlässlich sind, um ein unverzerrtes Verständnis der Sternentstehung zu erlangen. Rein kinematische Untersuchungen übersehen zwangsläufig einen signifikanten Teil stellarer Strukturen und führen somit zu einem unvollständigen Bild der Galaxienentwicklung.

Contents

1 Introduction

Understanding the formation and evolution of galaxies remains one of the central goals of modern astrophysics. Galaxies like the Milky Way are the product of billions of years of hierarchical structure formation, driven by gravitational collapse, gas accretion, star formation and feedback processes. In recent decades, both observational surveys and high-resolution cosmological simulations have significantly advanced our knowledge of these complex systems [1–3]. The synergy between data and theory now allows us to explore the substructure of galaxies in high detail, both in real observations and in simulated analogs. A key aspect of this substructure is the process of star formation, and the resulting clustering of young stellar populations.

This thesis investigates the dynamical and chemodynamical clustering of star-forming regions in a Milky Way-like galaxy, utilizing the ultra-high-resolution simulations from the NIHAO-UHD project [4]. These simulations capture the internal structure of disk galaxies at resolutions sufficient to study star-forming regions. Key objectives of this work are:

- To show the possibility of identifying star forming regions using clustering algorithms.
- To compare the properties of these identified stellar groups derived from different feature spaces, specifically contrasting dynamical tagging (feature space based on positions and velocities) to chemodynamical tagging (adding chemical abundances to the dynamical feature space) [5].
- To explore the implications of these findings not only on simulations, but on observational data.

First though, it is essential to review the prevailing cosmological model that describes how galaxies and the structures within them form and evolve.

1.1 Hierarchical Structure Formation in Galaxies

Within the Universe we detect a variety of astrophysical structures, ranging from single stars to extensive galaxy clusters. The formation and evolution of these structures are best explained within the framework of the Λ-Cold Dark Matter (Λ-CDM) cosmological model.

1.1.1 The Λ-CDM Model

The widely accepted theory of the Λ-CDM model is often referred to as the standard model of cosmology, and currently provides our most comprehensive understanding of the Universe's origin and evolution. It describes a universe that [6–9]

- Began with the Big Bang approximately 13.8 billion years ago and underwent a period of rapid expansion known as inflation.
- Is currently experiencing accelerated expansion driven by dark energy (represented by cosmological constant 'Λ'), which constitutes ∼68% of its total energy density.
- Additionally is composed of ∼5% baryonic matter and ∼27% dark matter.
- Is spatially flat, isotropic and homogeneous on large scales.
- Has a geometry described by the Theory of General Relativity.

According to this model, structure formation started from weak density fluctuations in the otherwise homogeneous very early Universe. These fluctuations were then amplified over cosmic time by gravity, leading to the complex cosmic web of structures observed today.

The model is robustly supported by a wealth of observational evidence. For instance, measurements of the cosmic microwave background by the WMAP satellite [10], which probes the Universe about 380,000 years after the Big Bang, were combined with data of the 2dFGRS [11] galaxy redshift survey. The goal was to validate the central claims of the standard cosmological model and more precisely determine the Universe's geometry and matter composition. The observations confirm that the initial density fluctuations were consistent with a Gaussian random field, as predicted by cosmic inflation, and that the Universe's energy is indeed dominated by dark energy [6, 7].

1.1.2 Structures in the current Universe

The mass distribution in a Λ-CDM universe has a topology that is often described as a "cosmic web", a vast network of structures (see Figure 1.1). The classification of these large-scale structures can be based on the number of positive eigenvalues of their deformation tensor (the Hessian of the gravitational potential). The existing structures are 0D knots, 1D filaments, 2D sheets, and 3D voids, which are underdense regions of space [12]. Within this web, particularly at the intersections (knots) of filaments , overdense regions of dark matter, known as haloes, became gravitationally self-bound shortly after inflation [7]. These dark matter haloes are where baryonic gas cools, condenses, and coalesces, eventually forming the structures we identify as galaxies. However, the hierarchical nature of cosmic structure formation dictates that these galaxies rarely remain isolated. Instead, they are often gravitationally bound into larger assemblies that are located in the densest regions of the cosmic web, and can be broadly categorized by their richness and mass: Galaxy groups typically comprise a few to ~ 100 galaxies, while galaxy clusters are larger systems containing from $\sim$ 100 to thousands of galaxies. Their local environment is shaped by their position within the cosmic web. In turn, these environmental properties significantly influence the evolution and internal characteristics of their member galaxies [14].

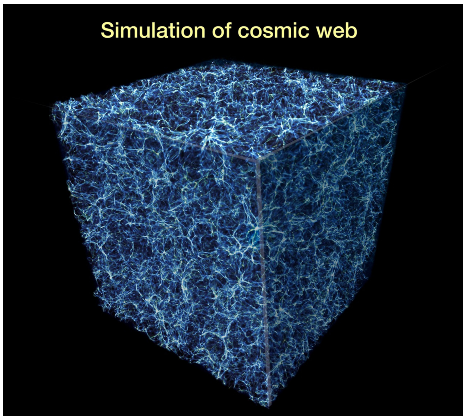


Figure 1.1: The "cosmic web" predicted by numerical simulations of structure formation in the universe. Galaxy clusters would correspond to fully white regions. Source: NASA [13]

These individual galaxies, whether found in relative isolation, within groups, or as members of massive clusters, have been observed throughout the Universe in various sizes (stellar masses typically $\sim 10^8 - 10^{12}\,\mathrm{M}_\odot$) and can be classified into four broad categories:

i. Elliptical galaxies: Characterized by smooth, nearly elliptical isophotes and predominantly older stellar populations.

ii. Spiral galaxies: Distinguished by thin disks with prominent spiral arm structures and a central bulge; these are sites of active star formation and thus host younger stellar populations.

iii. Lenticular galaxies: Intermediate forms possessing a smooth light distribution like ellipticals but also a thin disk and bulge, though typically without distinct spiral arms.

iv. Irregular galaxies: Lacking a dominant bulge or a rotationally symmetric disk, and exhibiting no obvious symmetry.

Since spiral- and barred spiral galaxies like the Milky Way have active star formation, we will focus on these types of galaxies. They typically consist of a thin, rotationally supported disk with spiral arms and a central bulge/bar component. The spiral structure is defined primarily by young stars, HII regions, molecular gas and dust absorption, contributing to their characteristically bluer appearance compared to elliptical galaxies (due to the prevalence of hot, young stars).

A central result about spirals is that stars and cold gas within the disk predominantly follow near-circular orbits in the disk plane. Therefore, the kinematics of a disk galaxy are largely described by its rotation curve $V_{rot}(R)$, that expresses the rotation velocity as function of galactocentric distance R. Crucially, observations reveal that $V_{rot}(R)$ often

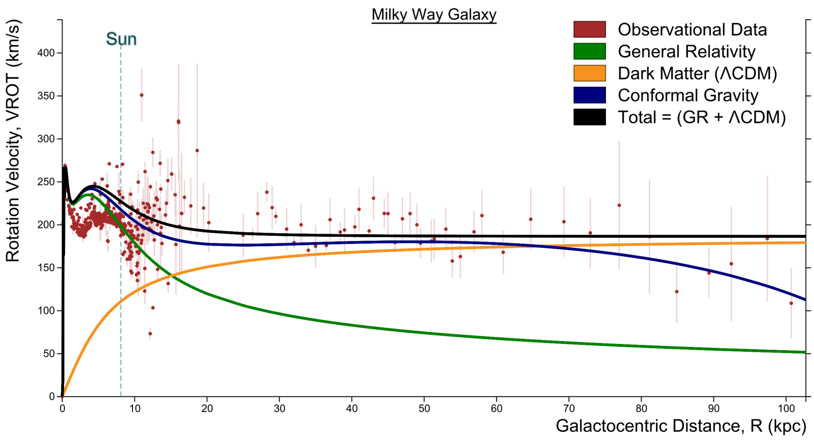


Figure 1.2: Different fits for the rotation curve of the Milky Way, according to different models. A combination of General Relativity and Λ-CDM gives the best fit to observational data. Source: [15]

remains approximately constant in the outer regions of galaxies (see Figure 1.2). This phenomenon is indicative of a significant amount of non-luminous 'dark matter' dominating the galaxy's total mass budget and shaping its overall gravitational potential [6]. This dark matter halo is a fundamental component of galaxies like the Milky Way and is crucial for the correctness of simulations like NIHAO-UHD.

The Milky Way itself serves as an important archetype for studying spiral galaxies and their star-forming processes. It is considered a relatively typical barred spiral galaxy (see Figure 1.3), with a thin stellar disk of mass $\sim 3.5 \times 10^{10}\,\mathrm{M}_\odot$, a radial scale length of $\sim 2.6\,\mathrm{kpc}$, a vertical scale height of $\sim 0.3\,\mathrm{kpc}$, going out to around $R \sim 60\,\mathrm{kpc}$ [16, 17]. The Milky Way also possesses a thick disk, comprising 10–20% of the thin disk's mass, with a vertical scale height of $\sim 1\,\mathrm{kpc}$, and slower rotation. The center of the Milky Way is known to harbor a supermassive black hole, Sagittarius A*, with a mass of approximately $4.2 \times 10^6\,\mathrm{M}_\odot$ [6, 16]. These properties are important to keep in mind when wanting to understand the characteristics of star formation within simulated galaxies analogous to the Milky Way.

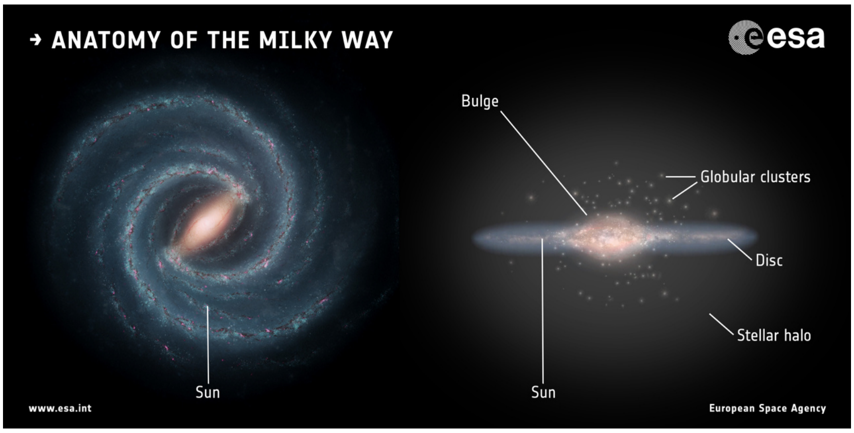


Figure 1.3: An artist's impression of the Milky Way as an example for a barred spiral galaxy. Image copyright: Left: NASA/JPL-Caltech; right: ESA; layout: ESA/ATG medialab [18]

1.2 Star Forming Regions

With this cosmological and galactic framework in place, it is now time to delve deeper into the nature of star-forming regions themselves. These regions are the places where stars are born, evolve in groups, and subsequently shape their host galaxies. They are typically located within the disks of spiral and irregular galaxies, particularly in the spiral arms where gas compression is heightened. Star formation happens in this multi-phase interstellar medium, where gas and cold temperatures give ideal conditions for stellar birth.

1.2.1 Structure of the ISM

The interstellar medium (ISM) refers to the matter and radiation that fill the space between stars in a galaxy, consisting primarily of gas (both ionized and neutral), dust particles, and cosmic rays. It exists in multiple thermally distinct phases which are regulated by turbulence and feedback processes [19].

In the Milky Way, about half the volume is occupied by a hot ionized medium with a number density $n < 0.01\,\mathrm{cm}^{-3}$ and kinetic temperature $T_K > 10^5\,\mathrm{K}$. The other half is filled by a combination of warm ionized medium and warm neutral medium with $n \sim 0.1 - 1\,\mathrm{cm}^{-3}$ and T_K of several thousand Kelvin. The neutral gas is subject to thermal instability that predicts segregation into the warm neutral medium (WNM) and cold neutral medium (CNM). The CNM, with $n > 10\,\mathrm{cm}^{-3}$ and $T_K < 100\,\mathrm{K}$ is where most of the star formation happens [19].

For this to initiate, gas in these regions must cool and condense further beyond the typical CNM state. This process begins when it cools within a time shorter than the age of the halo (efficient cooling), causing it to lose pressure support and being gravitationally pulled towards the center of the halo potential well. As density increases and temperature drops, the Jeans criterion ($M_{\mathrm{Gas}} > M_{\mathrm{Jeans}} \propto T^{\frac{3}{2}}\rho^{-\frac{1}{2}}$) is eventually satisfied, allowing gravitational collapse to dominate. Ultimately, this process may lead to the formation of dense, cold gas clouds. These clouds represent the coldest ($T_K \sim 10\,\mathrm{K}$ in absence of star formation) and densest ($n > 30\,\mathrm{cm}^{-3}$) part of space [20].

In the Milky Way and other nearby galaxies, it is found that all star formation takes place in these dense molecular clouds (e.g. [21, 22]). Additionally, observations of CO emissions from starburst galaxies show that their star-formation rates are also driven by the availability of dense molecular gas. Therefore, it is now generally accepted that the overall star-formation rate (SFR) of a galaxy is determined by its ability to form dense molecular clouds [6].

Giant Molecular Clouds

The largest dense molecular clouds considered to be single objects are giant molecular clouds (GMCs), with masses of $10^5 - 10^6\,\mathrm{M}_\odot$ and sizes of a few tens of parsecs. Within them, especially within the filaments, there may be small cores which are less massive but dense ($M \sim 10^2 - 10^4\,\mathrm{M}_\odot$ and $n_{H2} \sim 10^2 - 10^4\,\mathrm{cm}^{-3}$), as well as clumps with masses $M \sim 0.1 - 10\,\mathrm{M}_\odot$ and densities $n_{H2} > 10^5\,\mathrm{cm}^{-3}$. Clumps should be gravitationally bound and are considered to be star cluster forming regions. Cores on the other hand are regions where single stars are being formed [6]. This fragmentation naturally leads to the formation of stars in groups or clusters, rather than in isolation.

1.2.2 The Kennicutt-Schmidt Relation

Since the most obvious requirement for star formation to occur is the presence of gas, we expect there to be a relation between the surface density Σ_{SFR} of star forming regions, and the mean gas surface density Σ_{gas}.

$$\Sigma_{SFR} \propto (\Sigma_{gas})^N , \quad N \in \mathbb{R} \tag{1.1}$$

In other words, a relation between how much stellar mass forms per unit area per unit time, and how much gas exists per unit area. Such power-law relations between these two variables date back to Schmidt [23] and are therefore commonly referred to as Schmidt relations. Numerous observational studies have shown that the Schmidt relation is a good description of the global star-formation rates (averaged over the the star-forming disk), with the best values of N typically between 1 and 2.

Kennicutt [24] then included starburst galaxies in his analysis and found, that these systems follow the same power-law relation beween Σ_{SFR} and the total surface densities of gas (atomic and molecular) as normal spiral galaxies. The best fit to observational data gives

$$\Sigma_{SFR} = (2.5 \pm 0.7) \times 10^{-4} \left(\frac{\Sigma_{gas}}{\mathrm{M}_\odot \mathrm{pc}^{-2}} \right)^{1.4 \pm 0.15} \mathrm{M}_\odot \mathrm{yr}^{-1} \mathrm{kpc}^{-2} \tag{1.2}$$

This will be important later, as the Kennicutt-Schmidt relation is a standard tool used in the simulation of galaxies [6, 20].

1.2.3 Star Formation Feedback

A fundamental constraint on models of galaxy formation is the observed low efficiency with which baryons are converted into stars. Observations indicate that less than 10% of the global baryon budget today is in the form of stars. This is contrary to predictions of early CDM models without some sort of feedback. In these frameworks, gas is expected to cool efficiently within dark matter halos, collapse into dense structures, and be converted into stars by the present day. This discrepancy is solved by the inclusion of stellar feedback processes that regulate star formation. In these processes, stars, particularly

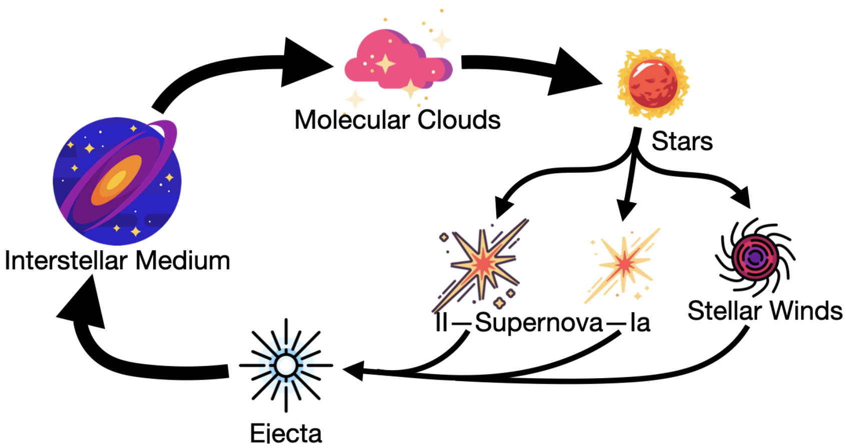


Figure 1.4: A simplified view of the baryonic cycle of star formation. Illustration by Berkay Guenes, adapted from [25]

the most massive ones, inject energy and momentum back into their surroundings. This can happen by driving powerful stellar winds, or when they produce Type II or Type Ia supernovae, as illustrated in the simplified schematic of the baryonic cycle (Figure 1.4) [8, 26]. The combined energy, momentum and chemically enriched material from these events are returned to the ISM. This injection of energy heats the surrounding gas, disrupts the molecular clouds and can drive large-scale galactic outflows that remove gas from the galaxy altogether.

Therefore, these processes effectively throttle star production, ensuring it remains an inefficient process over cosmic time, in agreement with observations. Simulating these feedback processes accurately is one of the main challenges in current simulational approaches.

1.2.4 The Inference of Stellar Ages

The inference of star ages is a foundational technique within Galactic archeology, the field dedicated to reconstructing the Milky Way's formation and evolutionary history by analyzing its oldest stars and stellar populations [27]. Identifying active or recent star-forming regions usually involves finding young stars. Furthermore, the chemical composition of stars carries imprints of their birth environment, allowing us to trace their origins.

Key to this is understanding stellar ages and metallicities. Metallicity describes the abundance of elements in an object that are heavier than hydrogen and helium. The abundance of heavy elements, like iron, in the ISM generally increases over cosmic time due to nucleosynthesis in stars and subsequent enrichment. Meanwhile, hydrogen is the primary primordial element, formed during the Big Bang. Thus, the [Fe/H] ratio in stars often serves as a marker for their formation epoch within a given galactic system.

The metallicity of a star is then defined as

$$\left[\frac{Fe}{H}\right] = \log_{10}\left(\frac{N_{Fe}}{N_H}\right)_* - \log_{10}\left(\frac{N_{Fe}}{N_H}\right)_\odot \tag{1.3}$$

where N_{Fe} and N_H are the number of iron and hydrogen atoms per unit of volume respectively, and * is the symbol for the star in question, comparing it to the metallicity of the sun [28].

Using this property, stars can then be classified into physical classes, such as populations I, II and III. These populations directly relate to their age, such that population I stars are the most recently formed, and population III stars are the first stars that formed right after the Big Bang. While population III stars are still a theoretical concept, population I and II stars are observed directly in the Milky Way. Importantly, due to the differing chemical abundances between stars born from differing gas clouds and of different types, the abundances of each metal within each star serve as a record of its formation and evolution and can be used to associate the stars with their parent structure [6].

1.3 Dynamical and Chemical Signatures of Star Clusters

A key aspect of this archeological approach is deconstructing galaxies into their fundamental building blocks: stars and the clusters in which many of them are born. Star clusters are thought to be the primary sites of star formation. However, over cosmic timescales, most of these clusters dissolve, dispersing their constituent stars throughout the galactic disk, halo or bulge. Identifying these dispersed stellar structures, or "tagging" them as co-natal, is a central goal in unraveling the galactic history. The challenge lies in the fact that, once dispersed, these stars are no longer spatially coherent, requiring more sophisticated methods to detect their shared past. Two ways have emerged for this purpose: Dynamical tagging, which leverages the conserved or slowly evolving kinematic properties of stars, and chemical tagging, which relies on the unique and largely immutable chemical abundances inherited from their natal molecular cloud (see 1.2.4). Both of these methods have been comprehensively reviewed by Helmi [29], and will be summarized in this chapter.

1.3.1 Dynamical Tagging

The fundamental premise of dynamical tagging is that stars born within the same cluster initially share very similar positions and velocities. As the cluster disperses, while its spatial coherence is lost, remnants of its kinematic coherence can persist in phase space for a significant period. These clusters can manifest as moving groups, stellar streams, or more subtle correlations in velocity space.

Dynamical tagging has proven effective in identifying relatively young to intermediate-age dispersed structures, such as various stellar streams discovered in the Gaia astronomical survey [29]. However, it also faces significant limitations, the most impactful of which is phase mixing. Over timescales of a few Galactic rotations, differential rotation and scattering by giant molecular clouds or spiral arms progressively erase kinematic coherence, making older dispersed clusters dynamically indistinguishable from the general field population [5]. Thus, the effectiveness of this method is largely restricted to younger structures in more relaxed galactic environments.

1.3.2 Chemical Tagging

Chemical tagging operates on a different principle: stars born from the same well-mixed giant molecular cloud should inherit nearly identical, detailed chemical compositions. This “chemical fingerprint” is largely preserved on the stellar surface throughout a star’s lifetime, and will therefore differ for stars born in different environments [30]. The feature space for chemical tagging is usually inherently high-dimensional, comprising the abundances of as many distinct chemical elements as can be precisely measured.

The primary strength of chemical tagging is its approximate time invariance. Because chemical abundances are persistent throughout the star’s age, this technique can identify the remnants of even the oldest, most dynamically dispersed star cluster [5].

Despite this, chemical tagging is not without its own challenges. It is observationally demanding, requiring high-resolution, high signal-to-noise spectroscopy for precise abundance determinations of many elements. Another fundamental limitation concerns the intrinsic chemical homogeneity of the ISM and the uniqueness of chemical signatures. If different molecular clouds forming stars at similar times and locations in the Galaxy possess very similar chemical compositions, distinguishing their dispersed members becomes difficult [31]. This suggests that the ISM, particularly in the young disk, may be too well-mixed for unique tagging of individual, small clusters based solely on chemistry.

It is now evident that dynamical and chemical tagging offer distinct, yet potentially complementary, windows into the past lives of stars. In the context of cosmological simulations, we possess the advantage of “perfect” knowledge of both, the true kinematic state and the intrinsic chemical abundances of all star particles, as well as their true birth origins. This allows for a rigorous assessment of the efficacy of different tagging techniques.

1.4 Galaxy Simulations

Numerical simulations have become an indispensable tool, offering a complementary approach to observations by providing a controlled environment to model the complex, non-linear physics governing galaxy formation and progression. With them we can also explore time evolutions, get a full three-dimensional view and test theories where direct observations are impossible or incomplete. Over the past years, advances in numerical methodologies and computing speed have enabled extraordinary progress in our ability to simulate the formation of structure within a CDM model (e.g. [7, 9]).

1.4.1 N-Body Methods

The formation and evolution of galaxies are fundamentally driven by the dynamics of dark matter, which constitutes the bulk of matter in the Universe. On cosmological scales, dark matter is treated as a collisionless fluid of non-interacting particles. Its evolution is described by the collisionless Boltzmann-equation

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \mathbf{v}\frac{\partial f}{\partial \mathbf{r}} - \frac{\partial \Phi}{\partial \mathbf{r}}\frac{\partial f}{\partial \mathbf{v}} = 0 \tag{1.4}$$

coupled to Poisson's equation

$$\nabla^2 \Phi = 4\pi G \int f d\mathbf{v} \tag{1.5}$$

The collisionless Boltzmann equation describes the evolution of the distribution function of dark matter, $f = f(\mathbf{r}, \mathbf{v}, t)$, under the influence of the collective gravitational potential, Φ, given by Poisson's equation [32].

To solve these equations and get the dynamics of the dark matter particles, we can use N-Body methods: We sample the distribution function f by an ensemble of N phase-space points $\mathbf{r}_i$, $\dot{\mathbf{r}}_i$, $i = 1, \ldots, N$ with masses m_i. The core computational challenge of N-body simulations is then to efficiently calculate the gravitational force on each particle due to all other $N - 1$ particles. A direct summation of these pairwise forces scales as $O(N^2)$, which is computationally prohibitive for the millions of particles used in modern simulations [32].

To overcome this, various hierarchical or approximation methods are employed. Tree codes, such as the Barnes-Hut algorithm [33] or methods based on k-D trees [34], are widely utilized. These algorithms organize particles into a hierarchical tree structure (e.g. an oct-tree or binary tree). The gravitational force from distant groups of particles is then approximated using multipole expansions (e.g. monopole or quadrupole terms) of the matter distribution within a tree node, rather than summing individual particle contributions. This reduces the computational cost significantly, typically to $O(N \log N)$ [34, 35].

Once the gravitational forces are determined, the particle trajectories are advanced in time using symplectic integration schemes, such as the Leapfrog integrator. These integrators are chosen for their long-term energy and momentum constervation properties, crucial for the cosmological timescales simulated [32]. While these N-body methods accurately describe the evolution of dark matter, simulating the formation of observable galaxies also requires modeling the behavior of baryonic matter.

1.4.2 Simulating Baryons using Lagrangian Methods and Smoothed Particle Hydrodynamics (SPH)

The visible components of galaxies are made of baryonic matter, initially hydrogen and helium, which condense to form stars. To understand this process, it's crucial to develop a numerical scheme that can simulate the complex behavior of this gas. Astrophysical gas in cosmological simulations is usually treated as inviscid ideal gas characterized by the ideal gas law

$$P_{th} = (\gamma - 1)\rho u \quad \text{or} \quad P_{th} = nk_B T \tag{1.6}$$

where P_{th} is the thermal pressure, T the temperature, n the particle number density, ρ the mass density, u the specific internal energy, and γ the adiabatic index. The dynamics of this gas are then governed by the Euler equations. They express the conservation of mass (1.7), momentum (1.8) and energy (1.9).

$$\frac{D\rho}{Dt} = -\rho \nabla \cdot \mathbf{v} \tag{1.7}$$

$$\frac{D\mathbf{v}}{Dt} = -\frac{1}{\rho}\nabla P + \mathbf{g} \tag{1.8}$$

$$\frac{Du}{Dt} = -\frac{P}{\rho}\nabla \cdot \mathbf{v} + \frac{1}{\rho}\Gamma - \frac{1}{\rho}\Lambda \tag{1.9}$$

Here ρ, $\mathbf{v}$, u are fluid density, velocity and internal energy respectively.
$D/Dt = \partial/\partial t + \mathbf{v} \cdot \nabla$ is the material derivative, $\mathbf{g}$ represents body forces such as gravity, and Γ and Λ represent heating and cooling terms. The total pressure P can in our case

refer to the thermal pressure P_{th} defined in (1.6), as other pressure sources (like magnetic or radiation pressure) are often not explicitly evolved in the primary hydrodynamical solver for the types of simulations discussed here [32, 36].

To numerically solve these equations for the gas dynamics, techniques like Smoothed Particle Hydrodynamics (SPH) are employed. SPH is a fully Lagrangian, mesh-free method, meaning the fluid is discretized into a set of "particles" (i.e. interpolation points), each representing a small parcel of fluid with a given mass and carrying properties like density, velocity and internal energy. The "smoothed" aspect of SPH refers to how continuous fluid properties are estimated from these discrete particles: At any point in space, or at the location of any SPH particle, a fluid quantity is calculated by performing a weighted average over the properties of all neighboring particles within a certain radius. This radius is known as the "smoothing length" h, and the weighting is determined by a "smoothing kernel" function W. The kernel is typically a radially symmetric function that peaks at the particle's center and smoothly drops to zero at a distance related to h (see Figure 1.5). The smoothing length itself is adaptive, often adjusted so that each particle has a roughly constant number of neighbours, ensuring a consistent level of smoothing across varying densities [32, 36]. Because the SPH particles themselves carry mass, mass conservation is inherently guaranteed. The Lagrangian nature of SPH also couples naturally with N-body methods for gravity. These characteristics make SPH a powerful tool for simulating the baryonic physics in cosmological structure formation.

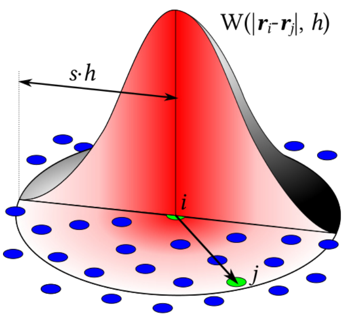


Figure 1.5: Schematic view of an SPH convolution (Figure reproduced from [37])

1.4.3 Extensions of the Models

The equations presented above provide a solid foundation, but they do not yet incorporate several key astrophysical phenomena, such as [32]:

- *Gas cooling*: Gas cools by dissipating internal energy through various processes like collisional excitation, metal line emission, and molecular cooling, which are modeled in simulations using cooling functions.

- *Star formation*: Star formation is modeled by converting eligible cold, dense gas (meeting criteria like density or Jeans instability) into collisionless star particles, typically using a probabilistic scheme based on a Kennicutt-Schmidt-like rate. These star particles represent single-metallicity stellar populations with an assumed initial mass function, and their subsequent modeled stellar evolution dictates mass and metal return to the surrounding gas.

- *Stellar feedback*: Stellar feedback is modeled by injecting energy and momentum from stars (primarily supernovae) into surrounding gas, either thermally (by temporarily disabling cooling or heating gas to overcome artificial overcooling) or kinetically (using hydrodynamically-decoupled wind particles for non-local impact). More explicit sub-resolution models now also incorporate other channels like stellar winds and radiation pressure, all designed to regulate star formation and launch galactic outflows by ejecting gas.

The implementation of these and other critical astrophysical phenomena into simulations has been a continued focus of research. This ongoing effort has recently resulted in simulations capable of producing increasingly realistic galaxies (e.g. [38, 39]). The NIHAO project [39], which forms the basis of our analysis, is one example of such advancements. We will revisit the specific implementation details of these phenomena within the NIHAO simulations in the next section.

2 Method of Analysis

2.1 The NIHAO-UHD Cosmological Hydrodynamical Simulations

This thesis utilizes simulations of Milky Way-like galaxies from the NIHAO-UHD (Numerical Investigation of a Hundred Astronomical Objects – Ultra High Definition) project [4]. These represent higher-resolution resimulations of galaxies from the original NIHAO suite [39], a set of 100 cosmological zoom-in hydrodynamical simulations. Zoom-in describes a technique focusing on a specific region of interest (e.g. the volume destined to form a galaxy halo). This volume is selected from a larger, low-resolution cosmological simulation and re-simulated at significantly higher resolution, only using the surroundings as context for initial conditions.

The simulations encompass haloes ranging from dwarf ($M_{halo} \sim 5 \times 10^9 \mathrm{M}_\odot$) to Milky Way ($M_{halo} \sim 2 \times 10^{12} \mathrm{M}_\odot$). They are run in a flat Λ-CDM cosmology using parameters from the Planck Collaboration et al. [40]. While retaining the same initial conditions as their parent NIHAO simulations, the NIHAO-UHD versions achieve a mass resolution increased by a factor of 8 to 16. This level of detail is essential for this work, as it allows for a more detailed characterization of individual star-forming regions.

For a detailed description of the simulation scheme, we refer to the NIHAO-UHD [4], NIHAO [39], and Gasoline2 [36] papers. Below, we provide a summary of how the simulations implement star formation and feedback mechanisms, as these aspects are most relevant to this thesis.

2.1.1 Star Formation in NIHAO-UHD

For our analysis, we're particularly interested in how the NIHAO-UHD simulations model star formation. This recipe follows Stinson et al. [41], so that we first apply criteria to determine which gas particles are eligible to form stars. Then we probabilistically determine which gas particles actually do so, such that on average the star formation rate (SFR) is similar to a Schmidt relation. These gas particles spawn a new star particle with a pre-determined mass, and the same velocity, position and metallicity as themselves, reducing their own mass by that same amount. Star particles can then add energy, mass and metals back to gas particles through feedback processes [41].

These criteria are implemented as follows in NIHAO-UHD:
Gas is eligible to form stars when it is dense ($n_{th} > 10.3\text{cm}^{-3}$) and cold ($T < 15,000\text{K}$), where the density threshold is set by

$$n_{th} = \frac{50 \cdot m_{gas}}{\varepsilon_{gas}^3} = 10.3\text{cm}^{-3} \tag{2.1}$$

using the gas particle mass m_{gas} and the gravitational softening of the gas ε_{gas}. The value of 50 denotes the number of neighboring particles that get considered in the calculation of the smoothed hydrodynamic parameters. The initial stellar particle mass at birth is set to $1/3 \cdot m_{gas}$ of the initial gas mass, and mass loss is set according to stellar evolution models [4].
Ideally, stars should form whenever these specific criteria are met, but because our simulations have limited resolution, we can only use a probabilistic approach to capture the global behaviour of star formation: The formation is assumed to proceed according to

$$\frac{d\rho_\star}{dt} = c^\star \frac{\rho_{gas}}{t_{form}} \tag{2.2}$$

where ρ_g is the density of the gas particle, t_{form} is the star formation time scale, and $c^\star$ is an constant efficiency parameter that can be set to match observations [41]. Then the probability of a gas particle forming a star particle is given by:

$$P = \frac{m_{gas}}{m_{star}} \cdot \left(1 - \exp^{-c^\star \Delta t / t_{form}}\right) \tag{2.3}$$

where $m_{star} = 1/3 \cdot m_{gas}$ of the initial gas particle, and Δt is the time step over which star formation is considered. A random number is drawn, and if it is less than P, a new star particle is created, inheriting the position, velocity and metallicity of its parent gas particle [41].

2.1.2 Stellar Feedback in NIHAO-UHD

Two modes of stellar feedback are implemented as described in Stinson et al. [42]. The first mode models the energy input from stellar winds and photoionization from luminous young stars. This happens before any supernovae explode, and is therefore referred to as "early stellar feedback". It consists of $\varepsilon_{ESF} = 13\%$ of the total stellar flux (2×10^{50} erg of thermal energy per $M_\odot$) of the entire stellar population being ejected from stars into surrounding area [4, 39].

The second mode of feedback models the energy input via supernovae type Ia and II. This is done by applying a delayed cooling formalism for particles inside the blast region. For type II supernovae, we look at the lifetime of the most massive stars, which is around 4 Myr. Then the feedback commences around this time after the formation of the star particle [4].

Feedback of type Ia supernovae is implemented as in Raiteri, Villata, and Navarro [43], so that energy and metals from SN Ia are distributed to neighboring gas particles via the SPH smoothing kernel [41].

One test for the employed feedback model is the baryon conversion efficiency $M_{\mathrm{star}}/M_{\mathrm{bar}}$. This describes the ratio of stellar mass to total baryonic mass. A galaxy with a value of one would turn all the available cosmic baryon budget into stars and indicate a failure of the employed feedback mechanism [4]. Buck et al. [4] find that the NIHAO-UHD simulations stellar mass growths are in good agreement with abundance matching results across cosmic times. They conclude that the simulations result in disk galaxies of realistic stellar masses [4].

2.2 Cluster Identification and Tracking: AstroLink and FuzzyCat

To extract and track star-forming clusters from the NIHAO-UHD galaxy simulations, we employ a two-stage pipeline utilizing the unsupervised clustering algorithms "AstroLink" [44] and "FuzzyCat" [45]. This pipeline, as described in Oliver and Buck [45], is designed for datasets whose representation evolves, such as in our case through temporal changes in simulated galaxies. First, AstroLink is applied to individual simulation snapshots to identify hierarchical cluster structures. Subsequently, FuzzyCat processes these AstroLink outputs to identify and track astrophysically and statistically significant "fuzzy" clusters, thereby encapsulating their underlying processes like temporal evolution.

2.2.1 AstroLink: Identifying Clusters in Snapshots

AstroLink is an astrophysical clustering algorithm designed to extract arbitrarily shaped, hierarchical structures from point-based data [44]. A key improvement over its predeces-

sor, CLUSTAR-ND [46], is its reliance on a single primary cluster extraction parameter, S. This parameter defines the lower statistical significance threshold for a group of points to be considered a genuine cluster, distinct from random density fluctuations.
AstroLink can automatically and reliably estimate an optimal S through a dynamical model-fitting process, minimizing user intervention and enhancing objectivity. The algorithm operates on the input particle data (P) from a single simulation snapshot through five main steps [44, 45]:

1. Data rescaling: If enabled (default), input data features are rescaled to have unit variance. This step standardizes the feature space, ensuring that all features can contribute more equally to the clustering process and preventing features with intrinsically larger variances or differing units from dominating the distance calculations.

2. Local Density Estimation: For each data point, the set of $k_{\rm den}$ nearest neighbours, $N_{k_{\rm den}}$ is found. The local density is then estimated using a multivariate Epanechnikov kernel [47] and a balloon estimator [48]. The logarithm of this density is then taken and rescaled to a range of [0,1]. This rescaled log-density, denoted $\log\hat{\rho}$, highlights relative density contrasts and normalizes the impact of noisy density fluctuations across the dataset.

3. Data aggregation: AstroLink constructs a hierarchy of overdensities by tracking connected components of data points. This is achieved by considering edges in a $k_{\rm link}$ nearest neighbour graph, with points processed in order of decreasing local density ($\log\hat{\rho}$). This process effectively builds an "aggregation tree", representing nested structures.

4. Model-Fitting: A statistical model is fit to a measure of "clusteredness" for all potential groupings derived from the aggregation tree. This model typically fits a beta distribution by minimizing the robust-to-outliers Wasserstein-1 distance between the data and the model. Based on this fit, each potential cluster's prominence is converted into a statistical significance. Only groups whose derived statistical significance S_g exceeds the automatically estimated (or user-defined) threshold S_{noise} are identified as statistically significant clusters. The automatic estimation of S is derived from this model-fitting stage.

5. Hierarchy correction: If enabled, the algorithm performs a correction step. For each significant cluster identified, the other branch formed at a merge point of the aggregation tree is also evaluated. If this branch also meets the significance criterion $S\sigma$ and satisfies conditions to prevent redundancy, it is also incorporated into the final hierarchy. This step ensures a more complete representation of significant structures at different levels of the hierarchy. It can for example reduce the depth of the resultant hierarchy, by eliminating clusters-within-clusters that only differ by a small number of points.

AstroLink's ability to identify statistically significant, arbitrarily shaped structures makes it particularly effective for finding diverse astrophysical structures, such as the remnants of infalling satellites and, for our purposes, potentially star-forming regions within the galactic disc and halo.

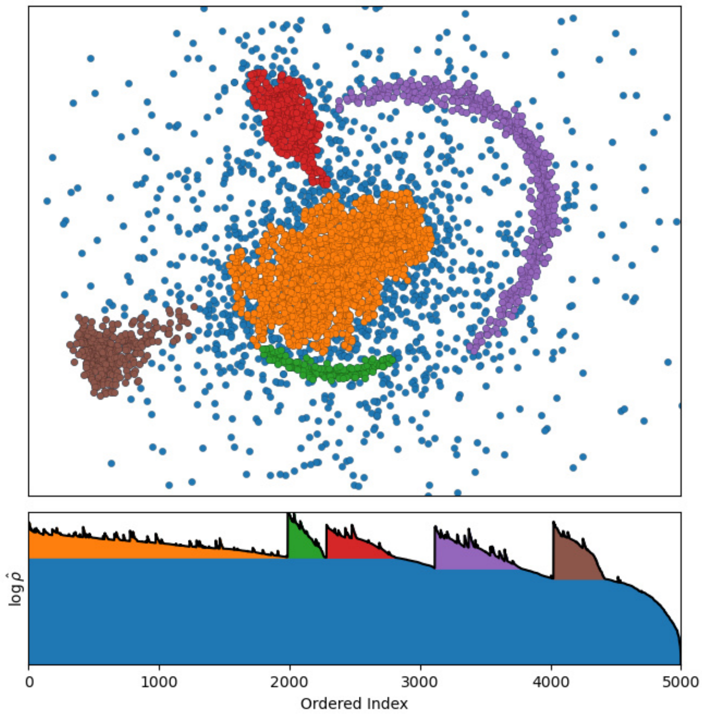


Figure 2.1: AstroLink applied to a toy-dataset, with detected clusters colored in red, green, brown and purple, and background noise in blue. The orange cluster is the largest statistically non-seperable overdensity in the data. The bottom panel represents the ordered-density plot, a 2D-representation of the clustering structure. Source: William Oliver [49]

2.2.2 FuzzyCat: Tracking Clusters across Snapshots

FuzzyCat is an unsupervised general-purpose soft-clustering algorithm designed to process a series of clusterings from an object-based data set to produce data-driven “fuzzy” clusters. It’s core function in this work is to identify groups of inter-snapshot clusters that exhibit temporal coherence, translating their evolution into fuzzy membership functions for the constituent particles. These membership functions encapsulate the effects of changes in cluster composition and characteristics due to the underlying processes, such as temporal evolution or statistical uncertainties. FuzzyCat is intentionally “content-agnostic”, meaning it makes no assumptions about the specific physical processes driving these changes, instead relying on the observed stability of particle groupings over time [45].

The operational mechanism of FuzzyCat shares conceptual similarities with AstroLink, particularly in its density estimation, aggregation and model-fitting stages (analogous to AstroLink’s steps 2, 3, and 4), but it operates on a dataset of clusters rather than individual data points. Key aspects of its functionality are [45]:

1. Jaccard-Space Representation: FuzzyCat first quantifies the similarity between every pair of input clusters (from any two snapshots) using the Jaccard index. This index measures the overlap in their particle membership, effectively transforming the problem into clustering in “Jaccard-space”, where each “point” is (in this work) an AstroLink-identified cluster and the “density” is related to the concentration of similar clusters.

2. Aggregation of Similar Clusters: Analogous to AstroLinks aggregation, FuzzyCat groups clusters that are highly similar (high Jaccard index). This process builds a hierarchy of “clusters of clusters”, representing potential tracked structures that maintain coherence across multiple snapshots.

3. Identification of Stable Fuzzy Clusters: From this hierarchy, FuzzyCat identifies the final tracked structures. This identification is governed by several key hyperparameters, like *minstability*, which defines the minimum duration over which a group of inter-snapshot clusters must maintain coherence to be considered a stable, tracked fuzzy cluster. The other hyperparameters are the internal similarity J_{min_intra} and the external

dissimilarity J_{max_inter}, which define the minimum Jaccard similarity required, as well as the maximum Jaccard similarity allowed among the constituent AstroLink clusters that form a single fuzzy clusters.
These conditions ensure statistical robustness and play a role analogous to AstroLink's "S" parameter in distinguishing significant, persistent structures from noise.

4. Fuzzy Membership Assignment: Once stable fuzzy clusters are identified, FuzzyCat translates them into membership functions for the underlying objects, which in our case are particles. For each particle and each tracked fuzzy cluster, the membership value typically represents the fraction of time that the particle was part of one of the constituent AstroLink clusters forming that fuzzy cluster.

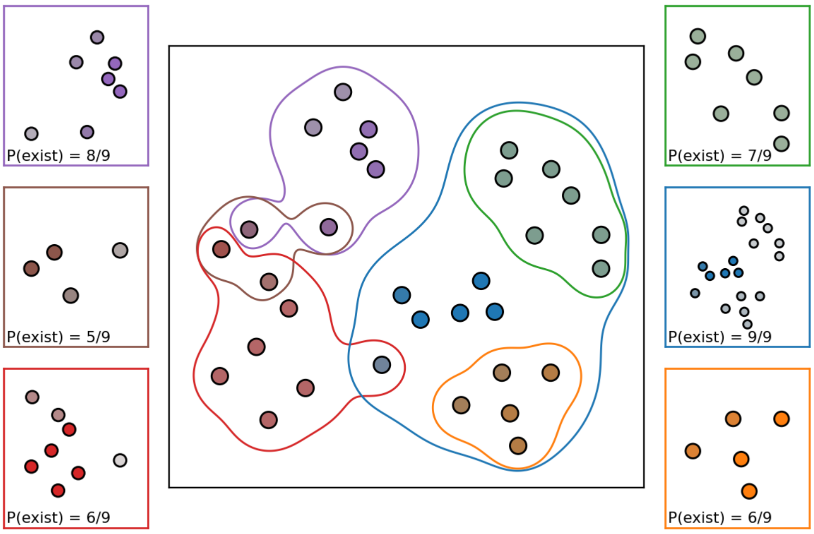


Figure 2.2: FuzzyCat applied to a toy-dataset. The central panel outlines the fuzzy clusters, with their colors representing how strongly they belong into each cluster. Each of the surrounding axes corresponds to one fuzzy cluster, and each point within them is colored according to it's membership to that fuzzy cluster alone. Source: William Oliver [50]

The application of FuzzyCat to the AstroLink-derived cluster catalogues is therefore crucial for overcoming the limitations of static structural identification. It enables the robust tracking of star-forming regions as they evolve within the simulation, providing a framework to study their continuity and transformation over cosmic time. Our specific strategy for configuring this AstroLink-FuzzyCat pipeline and interpreting the resulting tracked structures is the subject of the following chapter.

2.3 Cluster Identification and Classification Methodology

Our approach in this thesis is based on the AstroLink x FuzzyCat pipeline described in [45], as well as the code presented on the FuzzyCat documentation website [51]. We use the NIHAO-UHD galaxy simulation 2.79e12, and focus our analysis on the last 200 snapshots, which are equally-spaced time intervals. We implemented the ability to cluster on different particle types like “stars”, “gas”, “dark matter”, or a combination of them, but will focus primarily on stars. The feature space for their clustering in AstroLink can be chosen to be purely dynamical (six dimensions: positions (x, y, z) and velocities (v_x, v_y, v_z)), purely chemical (two dimensions: iron mass fraction ([Fe/H]) and oxygen mass fraction ([O/H])) or chemodynamical (eight dimensions: positions, velocities, iron and oxygen combined).

After we get the AstroLink clusters for every snapshot, we run FuzzyCat on them, to find stable fuzzy clusters in these 200 snapshots. In FuzzyCat, we calculate the *minstability* parameter such that the minimum life-span of a fuzzy cluster has to be at least 230 Myr, approximately the orbital period of our sun in the Milky Way.

With the fuzzy cluster results we can then commence our analysis of the star-forming regions, where the first step is to distinguish star-forming clusters from non-star-forming ones. This classification is done based on the age distribution of stellar particles within each cluster, analyzed at every snapshot of the simulation.

A cluster is identified as “star-forming” if it undergoes a rapid burst of star formation, during which it constitutes of at least 20 stars. From the implementation of the simulation, we know that a liberal estimate for the star burst time is around 50–100 Myr. We double this time, to account for the fuzzyness of the data, and therefore define a “burst event” by the criterion that the interquartile range (IQR) of a clusters stellar ages is less than $t_{75\_25} = 200\,\mathrm{Myr}$. The simulation snapshot where this condition is first met defines the cluster’s “birth snapshot”.

The full code for our analysis pipeline is available here.

3 Results of Analysis

3.1 Identification of Coherent Stellar Structures

The initial stage of our analysis involves applying the AstroLink algorithm to our chosen galaxy simulation. As established in Chapter 2.3, our focus is on the g2.79e12 galaxy from the NIHAO-UHD dataset. First, we perform a purely dynamical tagging analysis. Across the 200 simulation snapshots, AstroLink initially identifies approximately 40,000 candidate clusters. However, to ensure these structures are physically meaningful and persistent over time, we subsequently apply the FuzzyCat algorithm to this initial catalog. This yields 222 clusters coherent for ≥ 230 Myr.

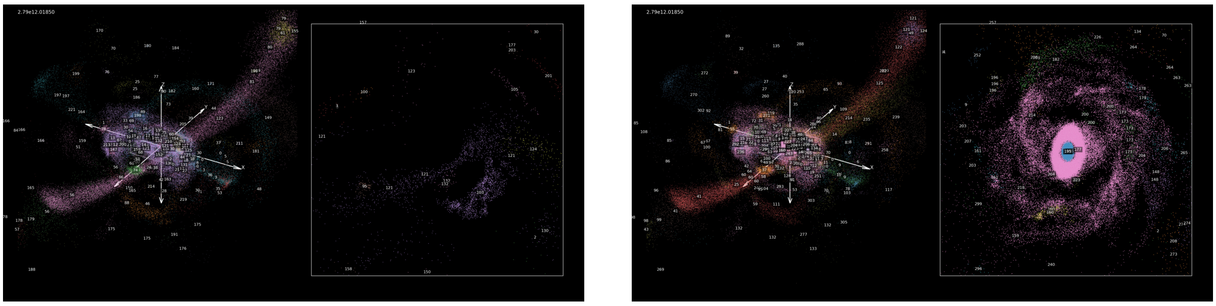
We then repeat the procedure using chemodynamical tagging, which reveals a greater degree of substructure, with AstroLink identifying approximately 55,000 clusters. After processing with FuzzyCat, this results in a final sample of 306 temporally coherent structures. By tracking the constituent particles of each identified cluster, we can characterise their structural and evolutionary properties throughout the simulation.

Figure 3.1 demonstrates the temporal coherence of our identified structures by tracking them across three snapshots spanning ~ 300 Myr. The color coding on the constituent particles and ID of a cluster reveals that clusters can be reliably tracked throughout the simulation.

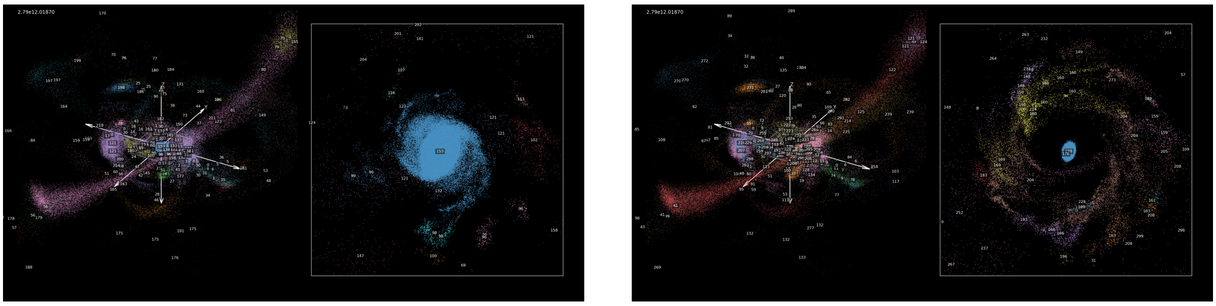
For the subsequent analysis, we first analyze the properties of the full sample of identified clusters before narrowing our focus to the sub-population of star-forming clusters. Every analysis will show the results of using dynamical tagging first, and then characterise how incorporating more information changes the clustering by including chemical abundances.

3.1.1 Spatial Distribution

To understand the location of these structures, we analyze their spatial distribution at the time of their first identification (Figure 3.2). We calculate the median r and $|z|$ over all star particles that are part of a cluster in this particular snapshot. Both methods primarily identify structures in the galactic halo, with distances far exceeding the stellar disk ($r_{\mathrm{med,dyn}} = 45.9\,\mathrm{kpc}$, $r_{\mathrm{med,chemdyn}} = 41.39\,\mathrm{kpc}$, $|z|_{\mathrm{med,dyn}} = 25.10$ kpc, $|z|_{\mathrm{med,chemdyn}} = 20.12$



(a) Snapshot 1850



(b) Snapshot 1870

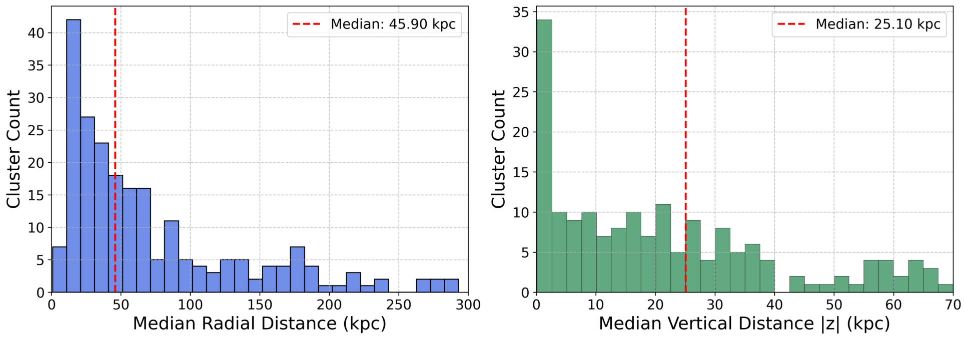
(c) Snapshot 1890

Figure 3.1: Fuzzycat clusters plotted in 3d-space and zoomed-in 2d disk view for different simulation snapshots (20 snapshots ~ 140 Myr) and tagging methods (dynamical left, chemodynamical right). Each color represents the constituent star particles of a distinct, temporally-coherent cluster identified by Fuzzycat. The zoom-in box spans 50×50 kpc.

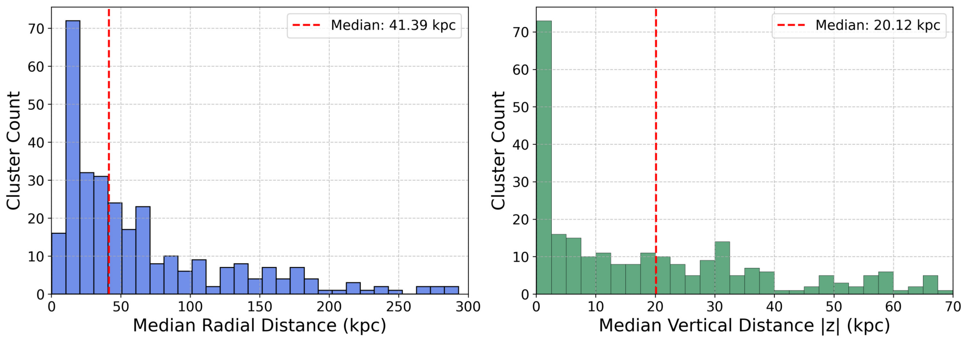
kpc). With the stellar disk extending to $r \sim 25$ kpc, these median values immediately reveal that our methods preferentially detect halo structures. The majority of clusters are found well outside the disk, extending to $r_{\max} \sim 300\,\text{kpc}$ and $|z|_{\max} \sim 100\,\text{kpc}$.

This halo-dominated detection is expected from hierarchical galaxy formation theory. The majority of these halo clusters most likely represent fossil remnants of accreted dwarf galaxies that have been tidally disrupted during hierarchical assembly. In the galactic halo, where the potential is smoother and tidal forces are weaker, these structures can maintain coherence over gigayear timescales.

A key distinction emerges between the two methods in the inner galaxy. Figure 3.2b



(a)



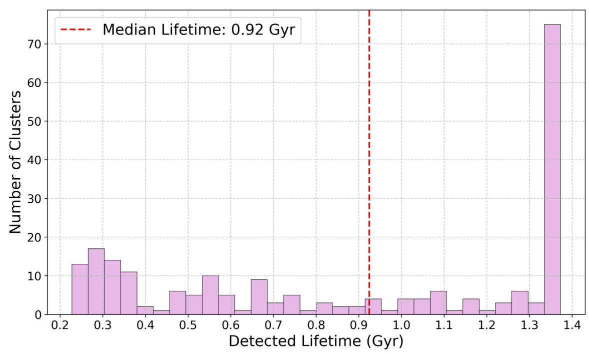
(b)

Figure 3.2: Spatial distribution of all fuzzy clusters identified using (a) dynamical tagging and (b) chemodynamical tagging. The galactic disk is approximately located at $r \lesssim 25\,\text{kpc}$ and $|z| \lesssim 5\,\text{kpc}$.

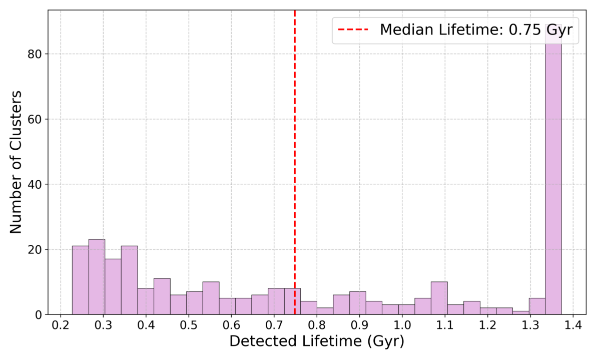
shows that chemodynamical tagging identifies a significantly larger number of clusters in regions with $r < 20$ kpc. This directly reflects the physics of these environments: The inner galaxy experiences strong tidal shear and differential rotation, which rapidly erases purely dynamical signatures on timescales down to ~ 50 Myr. This is much shorter than the cluster timescale of 230 Myr that FuzzyCat uses. By incorporating chemical abundances, which are conserved even as orbits diverge, chemodynamical tagging can recover structures that purely dynamical methods miss.

3.1.2 Observed Cluster Lifetimes

Having established where these clusters reside within the galaxy, we now examine their temporal evolution to understand how long they persist. Figure 3.3 presents the distribution of observed cluster lifetimes, representing the duration from initial detection until the cluster dissolves sufficiently to fall below our detection threshold. The distribution reveals that the median observed lifetime for chemodynamically tagged clusters is modestly shorter (0.75 Gyr) than their dynamically tagged counterparts (0.82 Gyr), although this is not a significant difference in the spread of the data (p-value of mood's test: 0.22). The lifetimes roughly match observed cluster disruption timescales for open clusters, with Boutloukos and Lamers [52] finding them to be around 1 Gyr in the neighbourhood of the sun ($r_{\rm sun} \approx 8$ kpc), and about 10 Gyr in the small magellanic cloud ($r_{\rm SMC} \approx 63$ kpc). Looking at the distribution, we do see that the inclusion of chemistry allows our algorithm



(a)



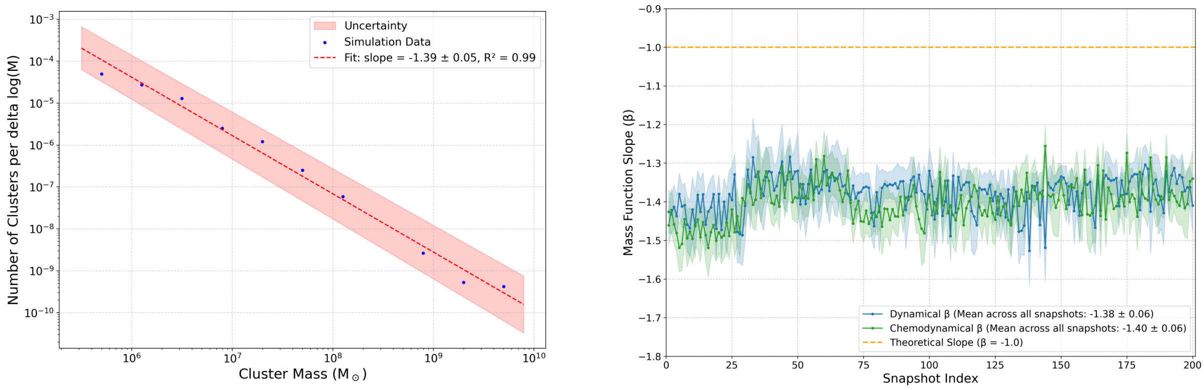
(b)

Figure 3.3: Distribution of observed cluster lifetimes for (a) dynamical tagging, (b) chemodynamical tagging.

to identify a distinct population of clusters that are inaccessible to a purely dynamical search. The dynamical signature of these additional short-lived clusters is comparatively weaker than those clusters that have been found using dynamics-only. As such, this typically means that they reside in the inner galaxy, where the dynamics are more heated, and therefore clusters are intrinsically shorter-lived. This would match our spatial analysis above, where the median radial distance of chemodynamical tagging was slightly lower than of purely dynamics, most likely because of this extra population of clusters. The prominent peak at the maximum lifetime of 1.38 Gyr represents a censoring effect. These are clusters that were present at the beginning of our analysis window and survived until the end, so their true lifetimes are likely even longer. They are most likely in the halo and resemble dwarf galaxies or streams.

3.1.3 Cluster Mass Function

The cluster mass function (CMF), which measures the number of clusters per mass interval (dN/dM), is a fundamental diagnostic of cluster formation and destruction processes. The CMF shape reflects the underlying physics governing cluster assembly, including fragmentation processes, feedback mechanisms, and environmental effects on cluster survival. Figure 3.4a shows the CMF for a single simulation snapshot. We bin the masses using the Freedman-Diaconis rule [53], plot $\log\left(\frac{dN}{d\log M}\right)$ against $\log(M)$ and perform linear regression to determine the power-law slope. The closest literature comparison for star clusters



(a) CMF of a single simulation snapshot (chemodynamical), demonstrating the linear fit over the mass range $10^5 - 10^{10}$ M$_\odot$. Shaded regions represent 1σ uncertainties.

(b) CMF over all snapshots for both tagging methods. Shaded regions indicate uncertainties of individual snapshots.

Figure 3.4: Cluster Mass Function Analysis for a) A single simulation snapshot and b) All snapshots

from theoretical models is a power-law where $\frac{dN}{dM} \propto M^{\alpha}$, with $\alpha \approx -2$ (eg. [54, 55]). Since we bin $d\log M$, this would mean $\frac{dN}{d\log M} \propto M^{\beta}$ with a slope $\beta = \alpha + 1 \approx -1$ for our analysis. For reasons we address in a later discussion, this value shouldn't be considered the ground truth in our case, with one of the concerns being that our clusters might not match the type of star clusters used in the literature analysis.

Our analysis yields slopes of $\beta \approx -1.38 \pm 0.06$ for dynamical, and $\beta \approx -1.4 \pm 0.06$ for chemodynamical tagging. These values are systematically steeper than the expectation, suggesting potential systematic biases in our cluster identification methodology. The steeper slope could indicate either an under-representation of high-mass clusters, or an overabundance of low-mass clusters due to fragmentation of larger structures. Analysis of the slope evolution across all snapshots (Figure 3.4b) reveals that this deviation remains relatively stable within ± 0.15 throughout the simulation timeline for both methods. The consistency of this offset suggests a systematic effect rather than stochastic variations.

3.1.4 Metallicity Distribution

The distribution of cluster metallicities in the first snapshot of a clusters lifetime is shown in Figure 3.5 and 3.6. It shows a bimodal distribution in both, the iron mass fraction, and the oxygen mass fraction, which is a signature of two-phase galaxy formation.

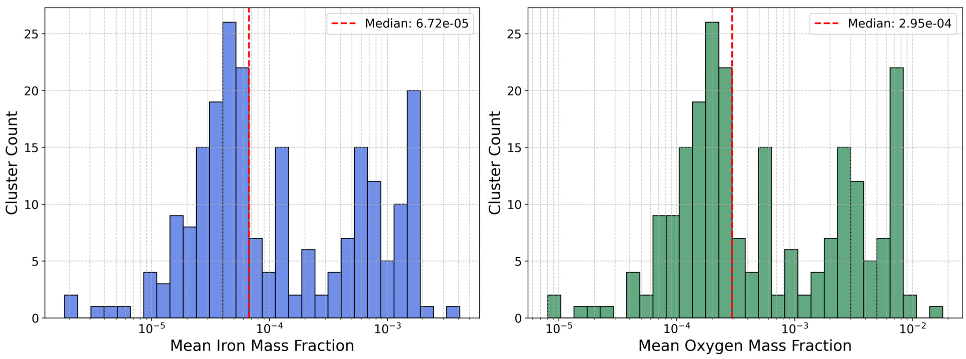


Figure 3.5: Metallicity distribution of all fuzzy clusters found in the galaxy using dynamical tagging. The metallicity value is calculated by taking the median metallicity over all particles in a cluster at the first snapshot of a clusters lifetime.

The first peak, at a low iron mass fraction of around 7×10^{-5}, represents the early assembly phase when metal-poor dwarf galaxies were accreted to build the proto-halo. The second peak at a higher iron mass fraction of around 10^{-3} represents the in-situ population. These are stars that formed later from progressively enriched gas within the main galaxy's potential well. This observation further demonstrates the advantage of

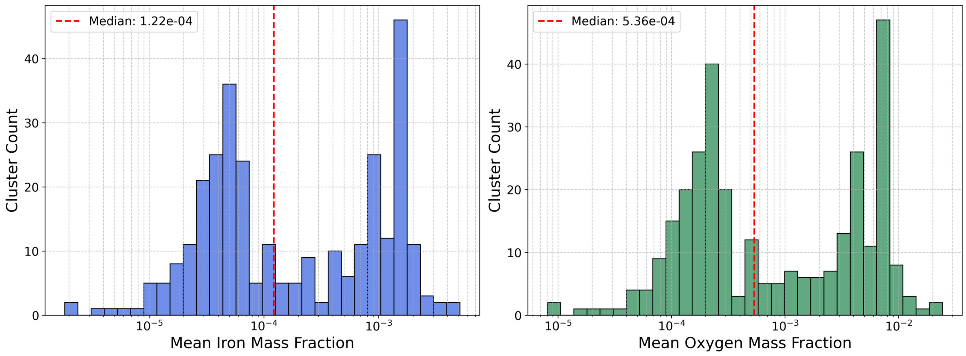


Figure 3.6: Metallicity distribution of all fuzzy clusters found in the galaxy using chemodynamical tagging. Calculation of metallicities is done as described in Figure 3.5.

chemodynamical tagging. In Figure 3.6, the two peaks are significantly more distinct and well-separated. The improved contrast demonstrates that including chemistry in the clustering process naturally groups stars with similar enrichment histories, leading to more chemically homogeneous clusters.

3.2 Star-Forming Clusters

We now focus on the subset of actively star-forming (SF) clusters, identified by their small age dispersion (< 0.2 Gyr) as detailed in Chapter 2.3. This selection is highly effective for identifying young stellar associations, providing insights into the immediate products of the star-formation process.

At the same time, the age dispersion criterion is, by definition, sensitive to any stellar population that formed rapidly, regardless of its absolute age. Consequently, in addition to the expected young disk clusters, our method also identifies a handful of outliers (about 10%). These objects are not sites of current star-formation, but rather "fossil records" of it. Their properties are consistent with those of globular clusters and the remnant nuclei of accreted dwarf galaxies.

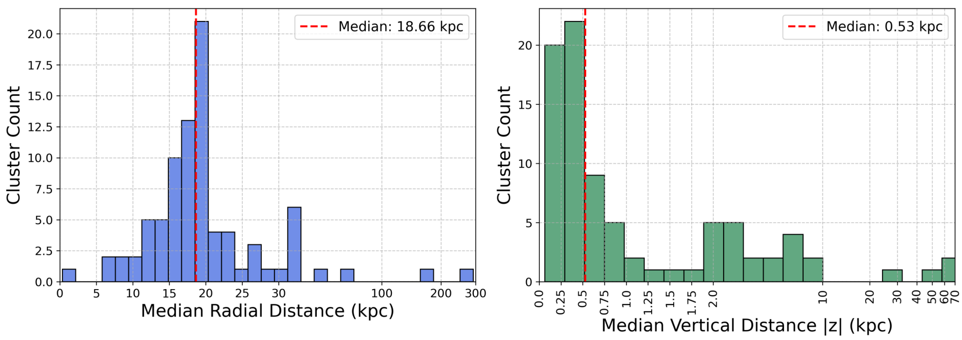
In all analyses, the two identification methods show a marked difference in their sensitivity to SF regions. Dynamical tagging identifies 38 out of 222 total clusters as star-forming (about 17%), while for chemodynamical tagging it is 85 out of 306 clusters (about 28%). Chemodynamical tagging therefore not only finds more clusters overall, but is nearly twice as effective at identifying star-forming regions.

This has significant implications for star-formation studies, as it suggests that previous dynamical-only surveys may have substantially underestimated the prevalence of young stellar associations, particularly in dynamically heated environments.

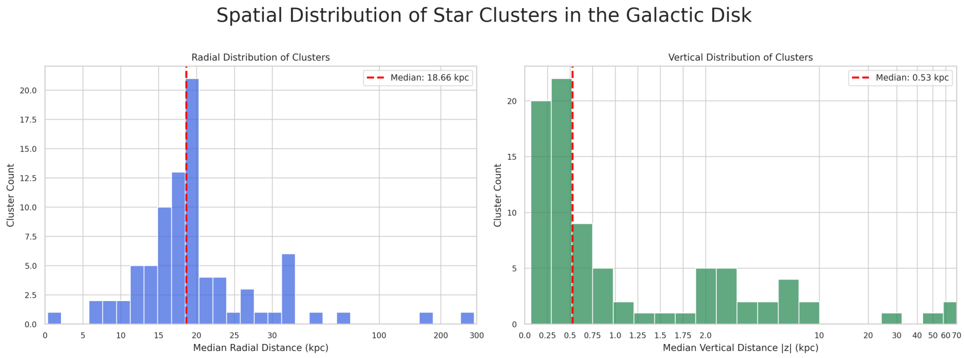
3.2.1 Spatial Distribution of Star-Formation

As expected, the spatial distribution of SF clusters is vastly different from the global population. Figure 3.7 shows that SF clusters are almost exclusively found within or near the galactic disk (e.g. $r_{\mathrm{med,dyn,SF}} = 18.6\,\mathrm{kpc} \Leftrightarrow r_{\mathrm{med,dyn}} = 45.9\,\mathrm{kpc}$), the primary site of ongoing star-formation in the galaxy. Some of the outliers are in the middle of the galactic halo, where we expect globular clusters, or at the edges of the galaxy, typical for an accreted dwarf galaxy. The comparison between methods is again revealing.

As discussed before, the galactic disk is a dynamically hostile environment, disrupting



(a)



(b)

Figure 3.7: Spatial distribution of SF clusters identified using (a) dynamical tagging and (b) chemodynamical tagging. The galactic disk is approximately located at $r \lesssim 25\,\mathrm{kpc}$ and $|z| \lesssim 5\,\mathrm{kpc}$.

young clusters on short timescales. Purely dynamical tagging struggles in this environment. So while dynamical tagging finds SF clusters in the disk, chemodynamical tagging identifies a much larger population, particularly in the inner disk. This confirms that even with modern clustering algorithms like Astrolink, purely dynamical tagging is still inferior for detecting star-formation in the most crowded and dynamically active regions of the galaxy.

3.2.2 Observed Cluster lifetimes

The lifetimes of SF clusters are by definition significantly shorter than the global population, with medians below 0.5 Gyr for both methods (Figure 3.8). The lifetimes are

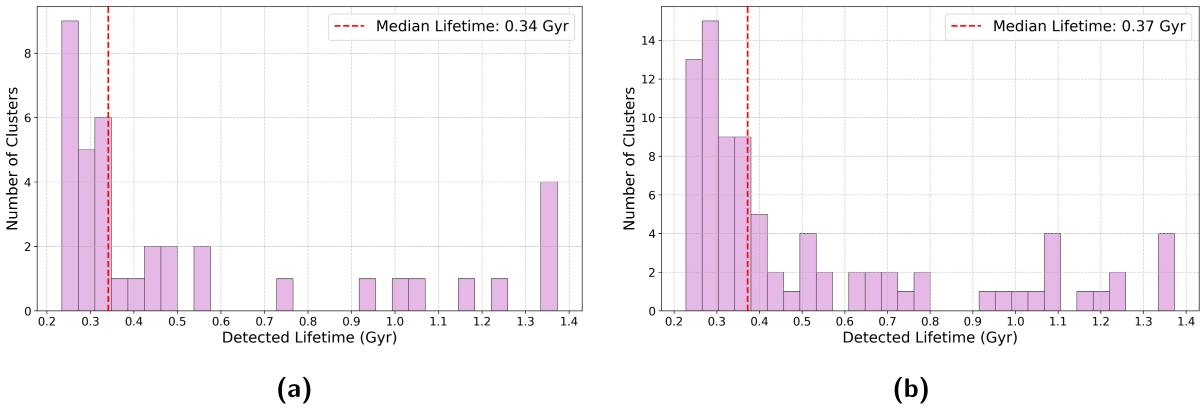


Figure 3.8: Lifetime distribution for (a) dynamical tagging, (b) chemodynamical tagging.

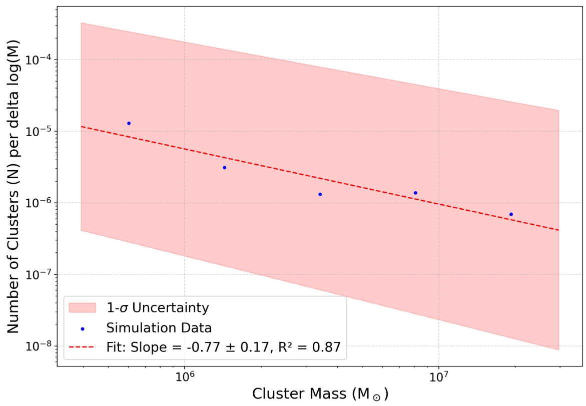
particularly revealing when compared to the galactic rotation period in the galactic disk (eg. ~ 230 Myr for the orbital period of the sun). This suggests that differential rotation and tidal shear disrupt most star-forming clusters within $1-2$ orbital periods, preventing them from surviving multiple passages through spiral arms or other density perturbations. The findings of Tarricq et al. [56] on a large survey of 389 local open clusters found in Gaia EDR3 data support this idea of cluster disruption by galactic tides on relatively short timescales.

The outliers with long observed lifetimes are, again, exactly what we expect from structures that have had one star-burst in the early stages of the universe, and have since been untouched by any tidal disruption.

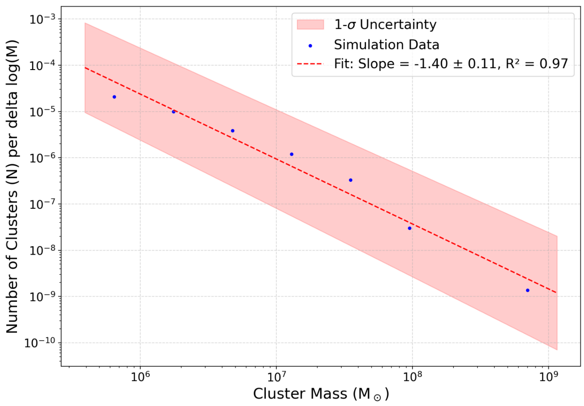
3.2.3 Cluster Mass Function

The reduced sample size for star-forming clusters introduces larger statistical uncertainties in the CMF analysis. To maximize our statistical robustness, we construct the CMF using the peak mass achieved by each star-forming cluster across its entire evolutionary history within our simulation timeframe.

This approach captures the clusters at their maximum extent while maintaining physical relevance to the star-formation process. The CMF reveals striking differences be-



(a) CMF for dynamical tagging with slope $\beta \approx -0.77 \pm 0.17$



(b) CMF for chemodynamical tagging with slope $\beta \approx -1.40 \pm 0.11$

Figure 3.9: CMF for star-forming clusters using (a) dynamical tagging and (b) chemodynamical tagging. Shaded regions show 1σ confidence intervals for the fitted slopes.

tween the two methods. Dynamical tagging yields a significantly shallower slope ($\beta = -0.77 \pm 0.17$) compared to theoretical expectations, suggesting a relative excess of high-mass star-forming clusters. In contrast, chemodynamical tagging produces a steeper slope ($\beta = -1.40 \pm 0.11$), with roughly the same discrepancy from the theoretical value of $\beta = -1$.

These results must be interpreted cautiously given the reduced sample sizes, which increase both statistical uncertainties and susceptibility to small-number effects. However, the shallow slope for dynamically tagged clusters may reflect the method's bias toward gravitationally bound, high-mass structures, potentially missing lower-mass star-forming associations that are more loosely bound. Conversely, the chemodynamical approach appears to capture a more complete mass spectrum of star-forming regions, including lower-mass associations identified through their chemical coherence.

3.2.4 Metallicity Distribution

The metallicity distribution for SF clusters (Figures 3.10 and 3.11) provides a direct probe of the galaxy's present-day chemical state. We find that these clusters almost exclusively

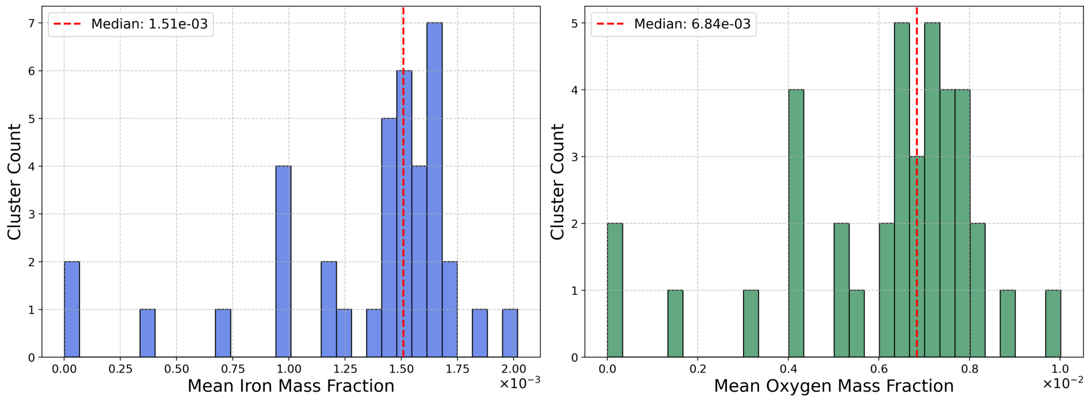


Figure 3.10: Metallicity distribution for dynamical tagging

populate the metal-rich mode seen in the global population, with a median iron mass fraction of around 1.5×10^{-3}.

This confirms that star-formation today is proceeding from a highly enriched interstellar medium. Outliers with lower metallicity fit the model of Globular clusters, which by definition will have a lower metallicity because they formed at an earlier time. The otherwise absence of metal-poor star-forming clusters implies that the era of significant metal-poor gas accretion has largely ceased in this galaxy at the observed epoch. Any gas that is currently being accreted must either be pre-enriched or is efficiently mixed with the disk's gas before it can form stars.

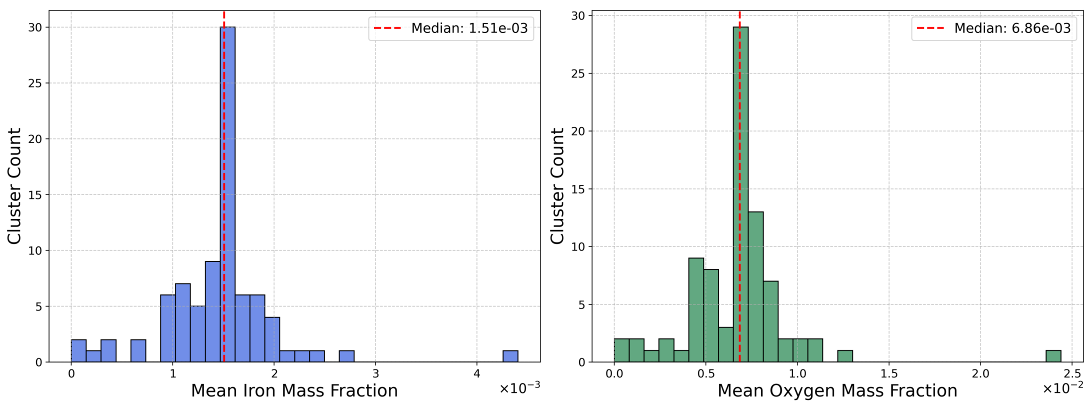


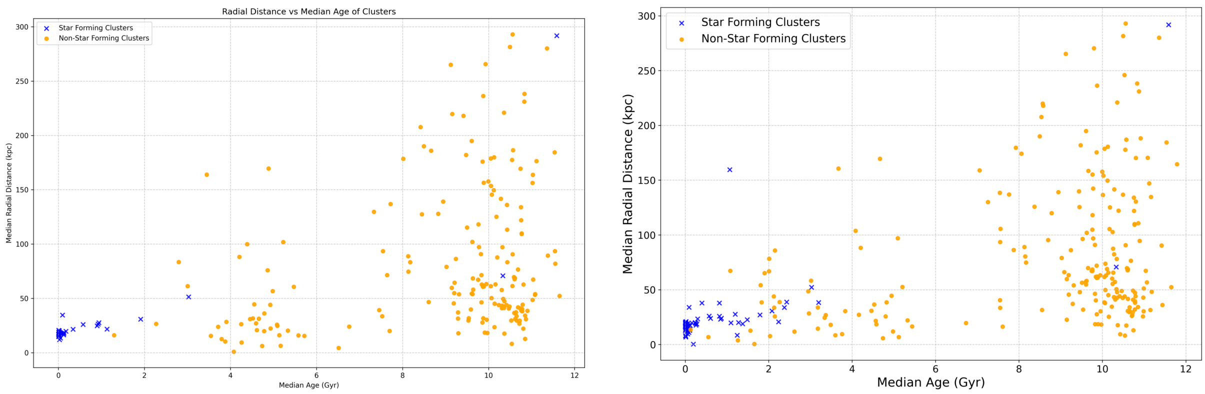
Figure 3.11: Metallicity distribution for chemodynamical tagging

3.3 Combining Statistics

By combining our derived statistics, we can construct a more complete evolutionary picture of the cluster populations.

3.3.1 Star-Forming and Non-Star-Forming Clusters

Figure 3.12 plots the median age of the star particles in a cluster at its detection against its median radial distance from the center, and differentiates between star-forming clusters and non-star-forming ones. It shows a clear trend: While there is a surprising amount of



(a) Dynamical tagging **(b)** Chemodynamical tagging

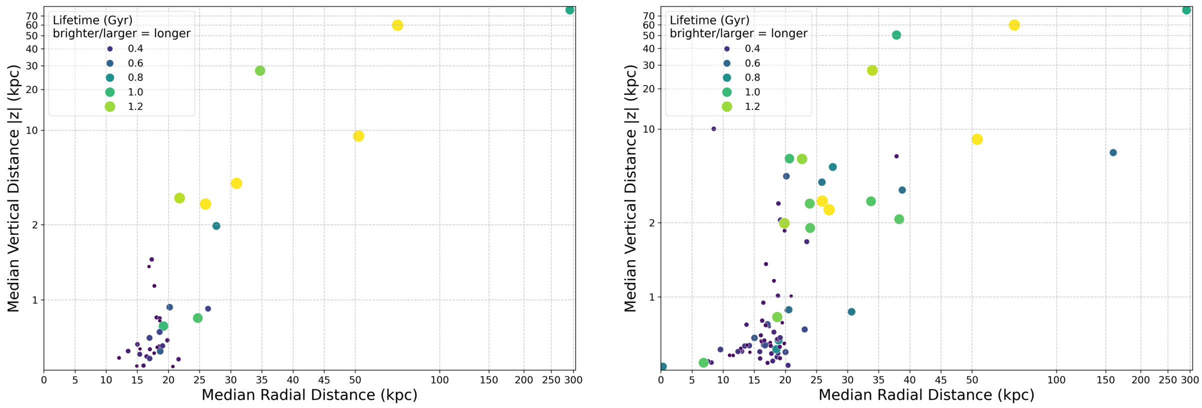
Figure 3.12: Median age of cluster constituents plotted against their median radial distance from the center. Blue markers represent star-forming cluster population.

rather old clusters in the disk, the further out a cluster is, the older it tends to be. Clusters younger than $\sim$ 2 Gyr, and therefore most star-forming clusters, are in-situ structures found within the inner $\sim$ 30 kpc.

In contrast, the ancient clusters (age $>$ 10 Gyr) are spread across the outer halo. This is a direct visualization of the “two-phase” galaxy formation process: The old halo formed first from external mergers, while the younger disk and inner halo populations were built up over time from internal gas.

3.3.2 Position and Lifetime

In the subsequent analysis, we combine each cluster’s spatial coordinates with its observed lifetime to investigate how environment influences cluster survival. Consistent with our earlier spatial study, we find that star-forming clusters predominantly occupy the galactic



(a) Dynamical tagging **(b)** Chemodynamical tagging

Figure 3.13: Cluster location in the galaxy, focusing on the inner regions. Color and size depict the observed lifetime of a cluster (brighter/larger ≡ longer lifetime).

disk, especially the outer regions. These are zones characterised by strong tidal perturbations, where they experience rapid disruption, leading to shorter lifetimes.

A handful of outliers residing well beyond the formal disk boundary, however, have extended lifetimes. The origin of these structures is not fully clear. They may represent globular clusters, young clusters formed in an accreted satellite galaxy that is now merging, clusters that were formed in the disk and then ejected into the less disruptive environment of the halo, or perhaps parts of the disk misclassified by AstroLink.

3.3.3 Contamination Analysis

Next, we investigate the contamination and loss of star-forming clusters. For each star-forming cluster, we go through the simulation snapshot by snapshot and classify an in-situ population I of stars (that increases over time) by checking if they are less than 100 Myrs old (generous estimate of star-burst time). We then calculate the contamination fraction as $CO = 1 - \frac{|I \cap C|}{|C|}$, with C being the full cluster population. It represents the fraction of stars that are in the cluster now, but are not part of the in-situ population.

The loss fraction represents the proportion of in-situ members that have escaped the cluster. We calculate it as $LO = 1 - \frac{|I \cap C|}{|I|}$. We also filter out the clusters that have no information available for a corresponding snapshot.

Figure 3.14 shows the contamination and loss for every cluster in the last snapshot that it could be detected, before dropping below the detectability threshold. Most investigated

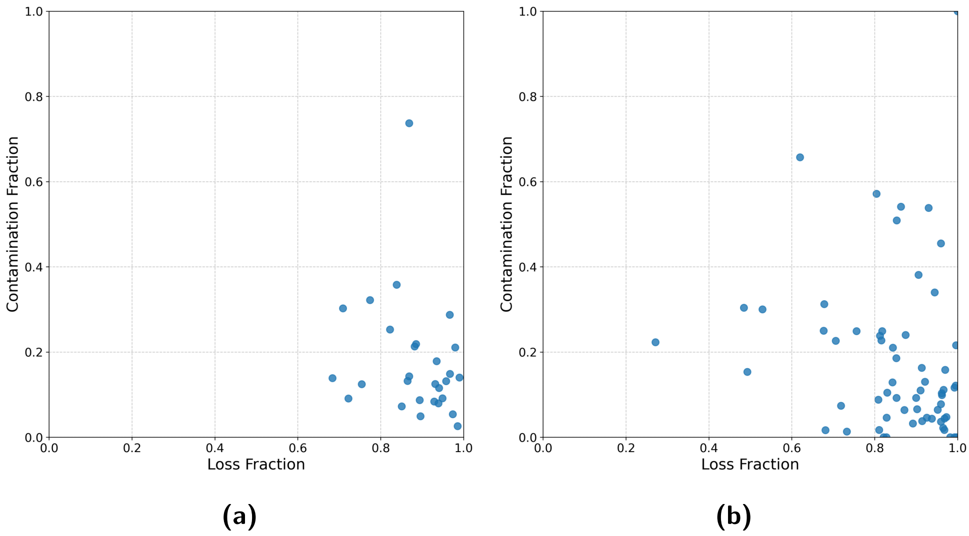


Figure 3.14: Loss and contamination fractions for a) dynamically tagged and b) chemodynamically tagged star-forming clusters.

clusters have contamination fractions below 0.2, and loss fractions above 0.7, meaning that throughout its lifetime, a cluster loses most of its original stellar content. At the same time, the cluster travels through the galactic disk and captures field stars, contributing to the contamination fraction.

We can then use these fractions to get a statistical handle on the probability that a given cluster found in the Milky Way today would have a particular loss/contamination. To do so, we marginalise over all snapshots as close to "today" as possible (the last 230 Myr of the simulation) and collect all contamination and loss fractions (around 200 data points for dynamical, around 400 for chemodynamical tagging).

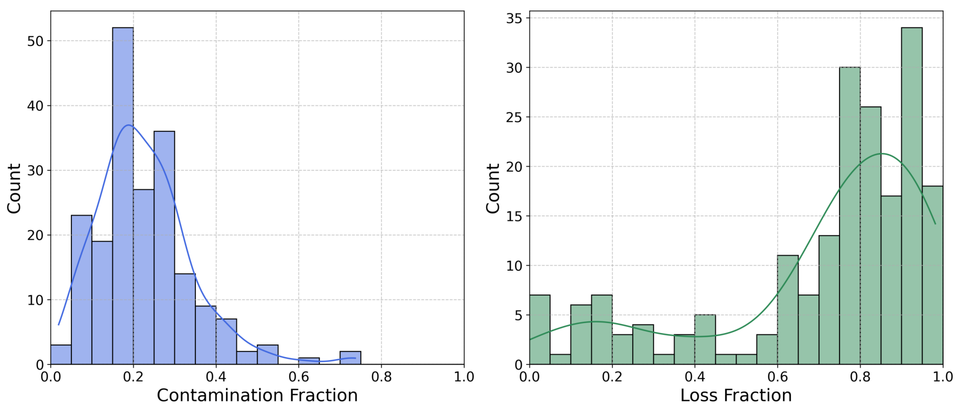


Figure 3.15: Histogram of all contamination and loss fractions collected for the last 230 Myr of the simulation using dynamical tagging.

The histograms for these are shown in figures 3.15 and 3.16. The contamination distribution peaks at around 0.2, while the loss has a small peak at around 0.3, and a larger peak at $0.8 - 1.0$.

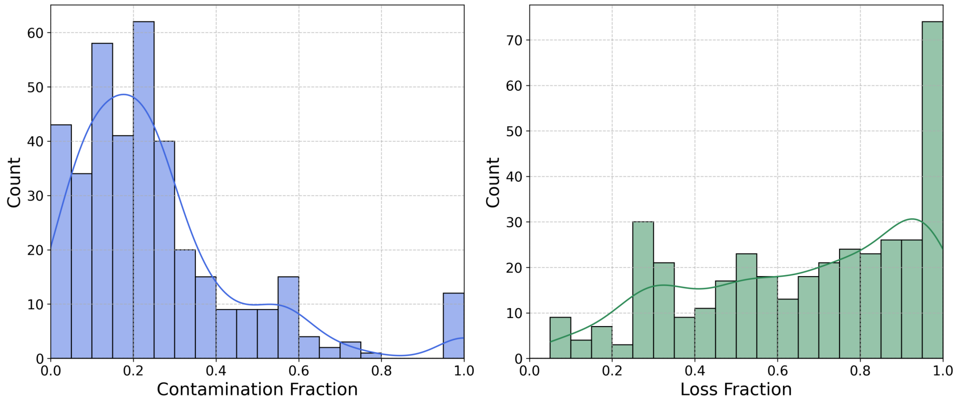


Figure 3.16: Histogram of all contamination and loss fractions collected for the last 230 Myr of the simulation using chemodynamical tagging.

Assuming sufficient correctness of our methods, this has huge implications for observations. First, it tells us that an observed cluster today is most likely to be moderately contaminated, with $10\% - 30\%$ of its stars being captured from the field rather than being born within it. Finding a perfectly pure or a completely contaminated cluster is less common. Secondly, regarding the retention of members born in-situ (loss), they are mostly dissolved in the background stars, having lost more than 80% of their original stellar population. The scarcity of clusters with intermediate loss fractions (e.g. 30%-50%) suggests that once a cluster starts to significantly unravel, it may dissolve very quickly and doesn't tend to remain in a half-lost state.

4 Discussion

Our analysis showed that statistical clustering algorithms can be used to find spatially, kinematically, and chemically coherent groups of stars in galactic simulations. The results presented reveal a significant divergence in the properties of stellar structures identified via dynamical versus chemodynamical tagging. The chemodynamical approach identifies $\sim 40\%$ more clusters overall and more than twice as many star-forming regions. Here we discuss the implications of these findings for both theoretical understanding and observational strategies.

4.1 Implications of Results

4.1.1 The Dominance of Chemodynamical Tagging

Throughout our analysis, chemodynamical tagging has consistently outperformed purely dynamical tagging across multiple metrics, especially when looking at star forming regions. While tidal forces and disk shearing destroy velocity coherence on timescales of ~ 50 Myr in the inner galaxy, stellar abundances remain unchanged. Stars that formed together retain their chemical fingerprint even after their orbits diverge. In regions of strong differential rotation, spiral arm passages, and GMC interactions, dynamical coherence is rapidly erased. Chemodynamical tagging recovers these “dynamically heated” populations that maintain spatial proximity and chemical homogeneity. However, precise chemical abundances can be harder to acquire than dynamical data, as it requires very high-resolution, high signal-to-noise spectroscopy over broad wavelengths to resolve weak absorption lines.

Nevertheless, finding double the amount of star-forming regions using chemodynamical tagging has deep consequences for our understanding of clustered star formation efficiency. If dynamical methods miss 50% of young associations, previous estimates of the fraction of stars forming in-situ (in the galaxy) may be severely underestimated. This is particularly acute in the inner galaxy ($r < 25$ kpc), where chemodynamical tagging reveals a rich population of star-forming structures invisible to dynamical methods.

4.1.2 The Cluster Mass Function

The systematic deviation of our CMF slopes from theoretical predictions provides insights into potential biases in our cluster detection methodologies. Both tagging methods yield slopes steeper than the canonical $\beta = -1$, indicating either a relative deficit of high-mass clusters, or equivalently a surplus of low-mass clusters compared to theoretical expectations. This deviation suggests a few possible explanations:

Misfit of theoretical value: The conditions under which the theoretical value of $\alpha = -2 \Leftrightarrow \beta = -1$ was found in e.g. Larsen, S. S. [55] differ significantly from those in our simulation. Their analysis focuses on star clusters with masses $< 10^5\,\mathrm{M}_\odot$, whereas our clusters begin at this mass threshold. Moreover, their sample is based on clusters identified chemically and subsequently verified for spatial coherence. As a result, their catalogue might omit clusters that have already dispersed in position space. All of the above reasons suggest that, for our specific simulation conditions and clusters, the CMF may still provide an appropriate fit.

Algorithmic fragmentation: The clustering algorithms may artificially fragment large structures into multiple smaller components. AstroLink, in particular, might preferentially identify substructure within larger assemblies, leading to an overabundance of low-mass detections. Future work could test this by comparing with alternative clustering algorithms such as HDBSCAN [57, 58] or by performing systematic completeness tests against a population of ground-truth star-forming clusters.

To investigate whether the bias originates from AstroLink or FuzzyCat, we calculated the CMF using all $\sim 55,000$ initial Astrolink detections before applying FuzzyCat. The persistence of the same systematic deviation in this larger sample suggests that the fragmentation, if present, occurs during the initial clustering phase rather than the temporal linking. Since low-mass associations are more susceptible to tidal disruption, and therefore temporally inconsistent, FuzzyCat might even be alleviating some of the discrepancy by removing more of the smaller clusters.

4.2 Robustness of the method

The reliability of our results depends on several methodological choices and inherent limitations:

Resolution and Completeness Limits: The NIHAO-UHD simulation's mass resolution ($m_\star \sim 10^4 \mathrm{M}_\odot$) sets a fundamental floor for cluster detection. This may contribute to the systematically different CMF slopes observed across both methods, but without multiple simulations at differing resolutions or alternative clustering algorithms for comparison, we cannot definitively distinguish their cause. However, the consistent deviation from canonical slopes suggests this is a systematic rather than stochastic effect.

Parameter Sensitivity: Our pipeline involves several key parameter choices, like the temporal coherence threshold of 230 Myr, the AstroLink significance parameter, the galaxy selection, and the criteria of a "star-forming" cluster. Alternative choices for these would likely lead to quantitatively different results (e.g. the exact number of clusters identified), however, the qualitative trends and the central conclusion are expected to be robust.

Statistical Power: Our sample sizes provide different levels of confidence across analyses: With the global population of 222 and 306 clusters tracked over 200 snapshots respectively, our statistical analyses are well-powered for identifying systematic trends and method comparisons. However, sampling only the star-forming clusters (38 and 85) reduces statistical reliability and limits precise parameter estimation, as reflected in the larger CMF slope uncertainties.

Despite these limitations, multiple independent factors suggest correctness of our analysis. The spatial distributions match theoretical expectations, the lifetimes of clusters match the environments we expect them to be in, and the chemodynamical approach works better than the dynamical across all metrics. The systematic differences between tagging methods suggest that they reflect genuine physical differences rather than methodological artifacts. While absolute values (eg. precise CMF slopes) should be interpreted cautiously, the relative comparisons and qualitative trends are robust. Future work incorporating alternative clustering algorithms, multiple simulation resolutions and parameter sensitivity analysis would further strengthen these findings.

4.3 Bridging Theory and Observation

Studying stellar clusters in a cosmological simulation provides unique advantages that are impossible to achieve observationally. We can track individual particles through cosmological time, perfectly distinguishing between a cluster's natal members and later contaminants. We have access to full 6D phase-space and detailed chemical abundances for every star, which rarely happens in observational astronomy, where only recently the newest Gaia Data Release 3 [2] now finally contains spectral data for around 220 million sources in the Milky Way. Our work leverages these advantages to draw conclusions for observational studies.

The results advocate for a shift in how observational surveys search for stellar structures, especially in the dynamically active regions of galactic disks. Purely kinematic searches, even with precise proper motions and radial velocities from surveys like Gaia, are fundamentally biased against the dynamically dispersed but chemically coherent associations that our work shows are prevalent.
Future observational campaigns should prioritize combining kinematic data with large-scale spectroscopic surveys, which will provide a far more complete census of stellar associations.
Using our contamination analysis as a basis for observational studies, we can also conclude that if an observer identifies a star-forming region in a galaxy like the Milky Way, our simulation predicts it is most likely a moderately contaminated system (contamination $\sim 20\%$) and is in the process of dissolution (loss $> 80\%$).

Ultimately, our work demonstrates that a complete census of stellar structures, from their birth in star-forming regions to their dissolution into the field, is only possible through the synergistic use of dynamics and chemistry. For both theorists and observers, relying on dynamics without chemicals provides a picture that is fundamentally incomplete.

5 Conclusion and Outlook

This thesis set out to investigate the nature of coherent stellar structures within a high-resolution, cosmological simulation of a Milky Way-like galaxy. We applied a pipeline using the AstroLink and FuzzyCat algorithms to identify and track stellar clusters over a ~ 1.4 Gyr period. The central goal was to systematically compare the effectiveness of a purely dynamical tagging method with a chemodynamical approach that incorporates stellar chemical abundances in the clustering.

We have shown that it is indeed possible to find distinct groups of stars with this method, from accreted dwarf galaxies to star forming regions. Our analysis reveals that chemodynamical tagging consistently provides a more complete and physically meaningful census of stellar structures. It identifies approximately 40% more clusters overall, and is twice as effective at locating young, star-forming clusters in the dynamically chaotic environment of the disk. Furthermore, the properties of these chemodynamically tagged clusters align more closely with theoretical expectations.

The primary conclusion of this work is that chemical information is not only complementary, but necessary for a full picture on star formation and galactic evolution. Relying on dynamics alone, especially in dynamically active environments like the galactic disk, will lead to a biased and incomplete view. The chemical fingerprint of a star-forming event is its most robust tracer, long after dynamical coherence has been erased.

The results and conclusions of this work offer several paths for future analysis:

- Investigate the results when only chemistry is used for tagging
- A follow up on the Cluster Mass Function by 1) applying HDBSCAN to the same dataset and calculating the CMF, 2) Doing a completeness analysis of AstroLink by injecting random "noise" clusters into the galaxy population
- Apply our analysis to other galaxies in the NIHAO-UHD dataset, and other high-resolution galactic datasets
- Trace the identified structures back to their progenitors (Gas in the Simulation) to investigate their physical origin

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